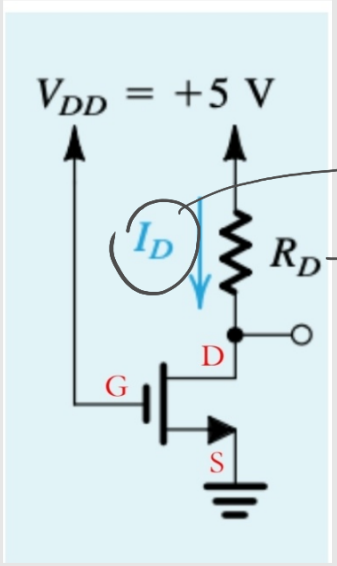
rd = 12kΩ Id = 0,4 mA VT = 1V



→ 0,4 mA

→ 12kΩ

lets check cut-off or not

$V_{GS} > V_t$

5V 1V

triode

$V_{DS} < V_{GS} - V_t$

4

saturation

$V_{DS} \geq V_{GS} - V_t$

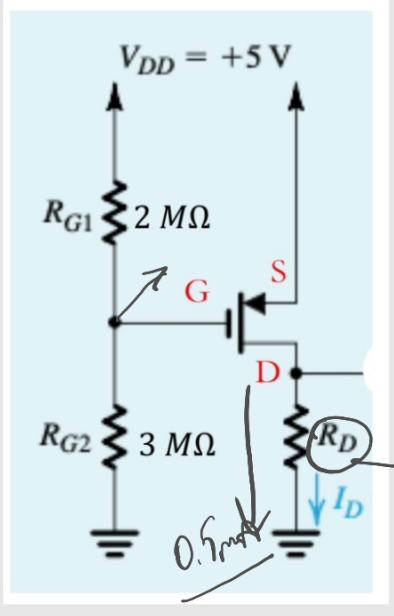
4

$V_D = 5 - 0,4mA \cdot 12k\Omega$

$5 - 4,8 = 0,2V$

0,2 < 4

triode



→ 0,6k

) Check the transistor operation mode in the circuit below $R_D = 6\ k\Omega, I_D = 0.5\ mA, |V_t| = 1\ V$.

its P-mos

$\frac{R_{G2}}{R_{G1}+R_{G2}} \cdot V_{DD} = V_G = 3V$

lets check cut-off or not

$V_{SG} > |V_t|$

$5-3 > 1$

triode

$V_{SD} < V_{SG} - |V_t|$

saturation

$5 - V_D \geq 2$

$5 - U_D$ $\quad U_S - U_G - U_t$

$5 - 3 - 1 = 1$

$U_D = 0{,}5\,mA \cdot 0{,}6k$

3V

SD saturation mode

if Rd is the edge of saturation ; Rd ?

$U_{SD} = U_{SG} - |U_t|$ (1)

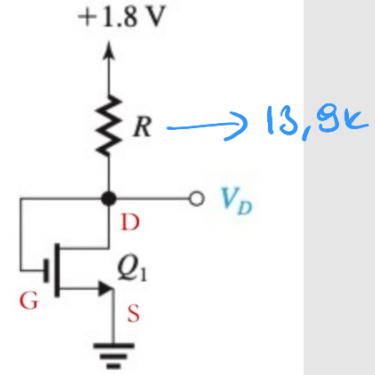
$\frac{U_S - U_D}{5}$ → H

$I_d = k \cdot (U_{SG} - U_t)^2$ at saturation → Its not depend on R

$0{,}5\,mA \cdot R = 4V$

8kΩ

3) Analyze the transistor circuit below. Take $R_D = 13.9\ k\Omega, V_t = 0.5\ V$.



$\mu_n C_{ox} = 0.4\ \text{mA/V}^2$

$\frac{W}{L} = \frac{0.72\ \mu m}{0.18\ \mu m}$

nmos → $U_D = U_G$

check for cut-off

U_{GS} ? U_t

$U_D - U_S$? U_t

Not

if UD and Ut connected each other it will be saturation

triode

saturation

$k_n = 0.8\ \text{mA/V}^2$

(*)

→ Solve for saturation

$$Id = 0{,}8 \cdot (U_G - 0{,}5)^2 = \frac{1{,}8 - U_G}{13{,}9\,k}$$

$$0{,}8 \left(U_G^2 - U_G + \frac{1}{4}\right) = \frac{1{,}8 - U_G}{13{,}9}$$

$$(0{,}8)\cdot(13{,}9)\left(x^2 - x + \frac{1}{4}\right) = 1{,}8 - x$$

$$11{,}10\left(x^2 - x + \frac{1}{4}\right) = 1{,}8 - x$$

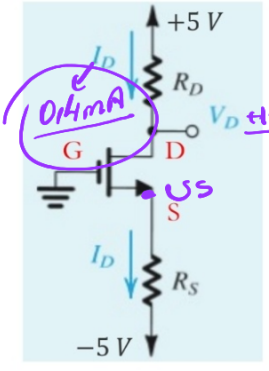
$$11{,}10\,x^2 - 11{,}10x + \frac{11{,}10}{4} - 1{,}8 + x = 0$$

$$11{,}10x^2 - 10{,}10x + 0{,}98 = 0$$

$$U_G = 0{,}8$$

check saturation

$$U_D - U_S \geq U_G - U_S - U_t$$



$\mu_n C_{ox} = 20\ \mu\text{A/V}^2$

$\frac{W}{L} = \frac{400\ \mu m}{1\ \mu m}$

$k_n = 0.4\ \text{mA/V}^2$

4) Design the transistor circuit below for $I_D = 0.4\ mA, V_D = +1\ V$. Take $V_t = 2\ V$.

$$k_n = \mu_n C_{ox} \cdot \frac{W}{L} \cdot \frac{1}{2}$$

check the cut-off

$U_G - U_S$? U_t

$$U_S - I_d \cdot R_S = -5$$

if current exist

triode

$V_{DS} < V_{GS} - V_t$

$I_d = k \cdot (2(V_{GS} - V_t) V_{DS} - (V_{DS})^2)$

saturation

$V_{DS} \geq V_{GS} - V_t$

$I_d = k \cdot (V_{GS} - V_t)^2$

$0.4\,mA = \frac{0.4\,mA}{V^2} \cdot V^2 \cdot (V_{GS} - 2)^2$

$V_{GS} = -1 \rightarrow V_S = 1$

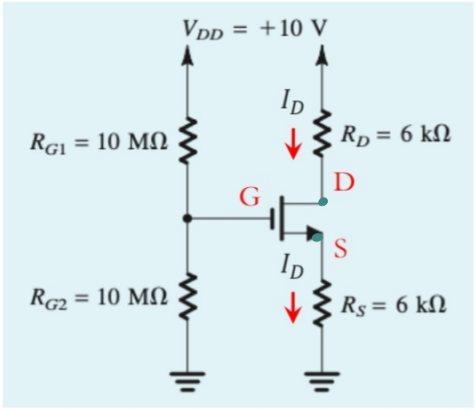
$V_{GS} = 3 \quad V_S = -3$

$-3 - 0{,}4mA \cdot R_S = -5$

$R_S = 5k$

$r_d = \frac{4}{0{,}4} = 10k$

5) Analyze the transistor circuit below. Take $V_t = 1\,V, k_n = 0.5\ \text{mA/V}^2$.



Check mode of operation

$V_G = 5\,V$

V_{DS} ? $V_{GS} - V_t$

$V_t = 1V \quad k_n = \frac{0{,}5\,mA}{V^2}$

$V_G = \frac{10}{20} \cdot 10 = 5V$

triode

$I_d = k \cdot (2(V_{GS} - V_t) V_{ds} - V_{ds}^2)$

saturation

$I_d = k \cdot (V_{GS} - V_t)^2$

$V_{GS} = 5 - 6I_d$

$10 - 6k \cdot I_d = V_d$

$10V = V_S + V_d$

$V_S = I_d \cdot 6k$

$V_{DS} = 10 - 12\,I_d$

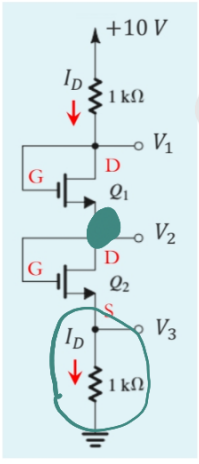
$I_d = \frac{0{,}5\,mA}{V^2} \cdot (V_{GS} - V_t)^2 \cdot V^2$

$I_d = 0.5mA\,(5 - 6kI_d - 1)^2$

$(4 - 6kI_d)^2$

$Id = 0,5\,mA$

8) Analyze the transistor circuit below. Take $|V_t| = 1\,V, k_n = 1\ \text{mA/V}^2, \lambda = 0$.



Check mode of operation

Both transistors in saturation

$V_D = V_G$

$U_3 - Id \cdot 1k = 0$

$U_3 = Id \cdot 1k$

$\frac{10 - U_1}{1k} = Id$

$U_3 = 10 - U_1$

$Id = k \cdot (U_{GS} - U_t)^2$

$Id = \frac{10-U_1}{1k} = \frac{U_3}{1k} = 1m \cdot (U_1 - U_2 - 1)^2 = 1m \cdot (U_2 - U_3 - 1)^2$

$U_3 = (U_2 - U_3 - 1)^2 \qquad U_3 = 10 - U_1$

$U_1 - U_2 - 1 = U_2 - U_3 - 1$

$2U_2 - U_1 = U_3$

$2U_2 - U_1 - 10 + U_1$

$U_2 = 5U$

$U_3 = (4 - U_3)^2$

$U^2 - 8U - 16 \qquad U^2 - 9U - 16$

3) The MOSFET in the circuit below has V_t = 1 V, k=1 mA/V² and λ=0. Given I_D=1.5 mA, analyze the circuit. Show all steps and calculations.

